Hyperspectral and Multispectral Image Fusion With Functional Data Analysis Techniques

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***Abstract*— The fusion of hyperspectral images (HSIs) and multispectral images (MSI) aims to synthesize high-resolution hyperspectral image (HR-HSI) from observable low-resolution hyperspectral image (LR-HSI) and high-resolution multispectral image (HR-MSI) data. Existing methodologies often overlook the importance of the degradation model within imaging systems and the inherent spectral and spatial smoothness properties of hyperspectral data, resulting in suboptimal outcomes in real-world applications. We introduce a novel approach that utilizes functional data analysis (FDA) techniques to precisely estimate degradation parameters and effectively fuse HSI-MSI pairs. Based on the physical generation mechanisms of both HSI and MSI, alongside the intrinsic smoothness of degradation parameters, we propose an iterative function estimation technique to recover the spectral response function (SRF) and point spread function (PSF). Additionally, we introduce a spatial content-aware spectral subspace regularizer (SSR) that promotes low-dimensional functional subspaces for spectral data within similar spatial contexts. Implemented within an alternating direction method of multipliers (ADMM) optimization framework, our fusion model integrates the precisely estimated degradation parameters and SSR to accurately construct the target HR-HSI data. Extensive experimental evaluations on multiple public datasets demonstrate that our approach achieves superior performance in both quantitative metrics and qualitative assessments. The code for this work is available at https://github.com/lingxingongye/FDAssr.git.**

***Index Terms*— Alternating direction method of multipliers (ADMM) optimization, degradation parameters, functional data analysis (FDA), HSI-MSI fusion, hyperspectral images (HSIs).**

NOMENCLATURE

N_W, N_H	HR-HSI width and height, respectively.
N_w, N_h	LR-HSI width and height, respectively.
N_S, N_s	Spectral dimension of HSI and MSI, respectively.
$\mathcal{Z}$	HR-HSI tensor.
$\mathcal{X}$	LR-HSI tensor.
$\mathcal{Y}$	HR-MSI tensor.
$\mathbf{Z}_{(3)}$	HR-HSI mode-3 matrix.
$\mathbf{X}_{(3)}$	LR-HSI mode-3 matrix.
$\mathbf{Y}_{(3)}$	HR-MSI mode-3 matrix.
$\mathbf{Q}$	Spatial degradation matrix.
$\mathbf{S}$	Spatial sampling matrix.
$\mathbf{R}$	Spectral degradation matrix.
r	Sampling factor.
i, j	Spatial indices of HS and MS images.
λ, β	Spectral index of HS and MS, respectively.
ϕ	Spectral response function (SRF).
Q	Point spread function (PSF).
u, v	Indices of PSF.
K	Number of iterations.
k	kth iteration.
$\widetilde{\mathcal{X}}$	Upsampled LR-HSI tensor.
L, l	Number and index of the basis of SRF.
$\boldsymbol{b}$	Coefficient vector of the basis of SRF.
φ	Spline basis.
M, N, m, n	Numbers and indices of the basis of PSF.
$\mathbf{C}$	Coefficient matrix of the basis of PSF.
T, t	Number and index of groups (superpixels).
$\mathbf{\Phi}$	Basis function matrix of HR-HSI.
$\bar{\mathbf{B}}_{(3)}$	Spline coefficients matrix of HR-HSI.
μ, ρ	Tuning parameters.
$\mathbf{G}_1, \mathbf{G}_2$	Lagrange multipliers.
ζ	Convergence threshold.

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I. INTRODUCTION

HYPERSPECTRAL optical imaging sensors simultaneously capture a wide range of spectral images, with wavelengths ranging from visible light to shortwave infrared [1], [2]. With its abundant spectral information, hyperspectral images (HSIs) enable precise materials identification, and have been successfully applied in fields such as remote sensing, medical diagnostics, and food safety [3], [4], [5]. Despite their inherent advantages, developing optical imaging systems faces constraints due to technological challenges, cost considerations, and signal-to-noise ratio limitations. This necessitates taking a balance between systems' spatial and spectral resolution to optimize performance [6], [7]. HSI often compromises spatial resolution to preserve more spectral information, producing low-resolution hyperspectral image (LR-HSI). Conversely, multispectral image (MSI) adopts a different trade-off, sacrificing spectral detail to achieve higher spatial resolution, producing high-resolution multispectral images (HR-MSI). By leveraging the complementary strengths of both LR-HSI and HR-MSI, it is efficient, economical, and practical to fuse these images from a shared target scene to generate high-resolution hyperspectral image (HR-HSI) [8], [9], [10].

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Although significant progress has been achieved in HSI-MSI fusion, several critical challenges remain unaddressed. First, most existing studies focus on the fusion algorithm itself, overlooking the importance of precisely modeling the imaging system's degradation process. LR-HSI and HR-MSI can be viewed as the degradation results of HR-HSI, and the essence of fusion is inverting the degradation process [11]. Therefore, accurate estimation of the degradation parameters is important. In the literature, datasets to train the fusion models are typically generated using user-specified degradation parameters. This approach implicitly assumes that the degradation parameters remain consistent across the training and testing stages for the fusion models. The assumption is certainly unrealistic in practical scenarios, where complex and variable conditions often affect the imaging system. As a result, neglecting accurate estimation and calibration of the degradation model from observational data can severely compromise fusion performance in real-world settings. Recognizing this, an emerging line of work employs diverse physical insights to improve degradation modeling [12], [13], [14], [15]. However, the literature still lacks a data-driven and robust way to estimate the degradation parameters from the observed data, particularly when the signal-to-noise ratio is relatively low.

As parameters of the degradation models, the SRF and PSF inherently are two smooth functions. Leveraging their smoothness property can effectively enhance both the accuracy and robustness of their estimation. Functional data analysis (FDA) [16], [17] provides a powerful framework for modeling such properties. Within this framework, we propose a novel iterative algorithm for estimating degradation parameters. Our approach demonstrates robustness and competitiveness, particularly in challenging scenarios with low signal-to-noise ratios. Furthermore, through comprehensive experiments, we show that the degradation parameters estimated by our method not only improve estimation accuracy but also benefit the subsequent recovery of HR-HSI data.

In addition to degradation parameter estimation, accurately constructing HR-HSI data remains to be challenging. Current fusion algorithms often seek to enhance performance by incorporating prior knowledge, typically drawn from natural image priors [6], [7], [18]. However, these approaches often neglect the distinct physical and spectral properties of hyperspectral data. HSIs exhibit smoothness along the spectral dimension and local smoothness in the spatial dimension. Despite being digitally recorded in a high-dimensional format, their intrinsic dimension is relatively low. FDA also emerges as an effective framework for capturing these characteristics. Building on this foundation, our second major contribution is the adaptation of the fusion algorithm within the FDA framework to recover hyperspectral information more effectively.

Moreover, for the fusion problem, many existing methods rely on assumptions such as global low rank, sparsity of spatial gradients, or other structural regularities. Some studies impose global low-rank constraints on the entire dataset, while others enforce local low-rank properties by partitioning the HSI into spatial blocks [19]. Recent research [20], [21], [22] highlights another key characteristic of hyperspectral data: spatial nonlocal self-similarity, which captures the recurrence of spatial structures across distant regions. Techniques such as non-local block clustering are commonly employed to exploit these patterns and improve the estimation process. However, the quality of spatial clustering plays a pivotal role in the accuracy of HR-HSI recovery. Homogeneous spectral content within spatial blocks often results in better fusion performance. Inspired by these observations, our third contribution is to introduce a concise yet highly effective prior for fusion models: the spatial content-aware spectral subspace regularizer (SSR). We employ superpixel segmentation [23] for detecting spectrally purer subgroups based on spatial content. Our model regularizer then promotes a low-dimensional functional subspace for the spectral data within each group. Our proposed model more informative prior for HR-HSI recovery.

Overall, the main contributions of this work are presented as follows.

1) We develop a novel iterative algorithm for estimating degradation parameters in FDA framework, leveraging the smoothness properties of the parameters to enhance estimation accuracy and robustness, particularly under low signal-to-noise ratio conditions.
2) We also construct a fusion model within the FDA framework, effectively capturing the spectral smoothness and spatial structures of hyperspectral data, leading to improved HSI recovery and fusion performance.
3) We propose the spatial content-aware SSR, which extracts spectrally pure subgroups and encourages low-dimensional functional subspaces for more efficient fusion reconstruction.

Through extensive simulation studies and real data analysis, we demonstrate that the proposed fusion method exhibits significant advantages, underscoring its effectiveness in addressing longstanding challenges in hyperspectral fusion.

The remainder of this article is organized as follows. Section II reviews related HSI-MSI fusion methods. Sections III and IV detail the proposed approaches for degradation parameter estimation and the fusion algorithm, respectively. Section V demonstrates the effectiveness of our method through quantitative and qualitative evaluations. Section VI provides an ablation analysis, and Section VII concludes this article with future research directions. Nomenclature summarizes the notations that will be used throughout the work.

II. Related Works

Recently, researchers have shown increasing interest in the HSI-MSI fusion problem and have proposed a variety of methods to address it. We categorize HSI-MSI fusion methods into three types: pansharpening-based methods, model-based methods, and convolutional neural network (CNN)-based methods. This section introduces each category in detail and ends with a review of the degradation model, a critical component of HSI-MSI fusion.

A. Pansharpening-Based Methods

Pansharpening [24] was first introduced to address the fusion problem between multispectral image and panchromatic image. The HSI-MSI fusion problem shares a fundamental similarity with pansharpening, as both aim to combine spatial and spectral information. Pansharpening methods, such as injection model [25], can be extended to HSI-MSI fusion by adapting their frameworks to handle the higher spectral

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dimensionality of HSI. For instance, spatial information from MSI can be injected into HSI while preserving its spectral integrity, enabling the application of traditional pansharpening techniques with appropriate modifications. Generally, pansharpening-based methods are computationally simple and efficient. However, they have limited modeling capabilities because they do not consider the spectral dimensional information of HSI, leading to poorer fusion accuracy.

B. CNN-Based Methods

In recent years, with the rapid development of deep learning, data-driven CNNs have achieved superior performance in the HSI-MSI fusion problem [26], [27], [28]. CNN-based methods learn the nonlinear mapping relationship from the observed data (LR-HSI and HR-MSI) to the target HR-HSI through CNN [29], [30]. Considering the complex nonlinear relationships between data, Dian et al. [31] used the Sylvester equation to obtain an initial estimate of the ideal HSI and then applied a CNN to learn the nonlinear mapping between the initial estimate and the reference image. Hu et al. [32] integrated the recently popular pixelShuffle [33] and attention modules to extract high-order features. Additionally, several CNN-based methods introduced degradation models into neural networks to enhance physical interpretability [34], [35]. For example, Dong et al. [36] combined degradation models with image priors and used end-to-end neural networks to learn the iterative optimization process. Huang et al. [37] employed unfolded neural network iterations to construct high-resolution HSI. However, these methods heavily rely on the "scale invariance" assumption. To overcome this limitation, Gao et al. [38] combined degradation models with attention modules and proposed an unsupervised strategy. Li et al. [39] proposed a two-stage neural network architecture to separately learn the nonlinear mappings of the degradation process and the fusion process.

Although CNN-based methods have demonstrated excellent performance in addressing the HSI-MSI fusion problem, they still face three inherent limitations. First, deep learning frameworks require a substantial amount of training data to effectively learn network parameters. However, acquiring such data is challenging in practice, particularly because training triplets (LR-HSI, HR-MSI, HR-HSI) from different satellites are limited. Second, the multiplatform, multienvironment, multiscale, and multitemporal characteristics of remote sensing satellite data introduce significant variations between datasets, which undermines the generalization capability of trained models. For instance, differences in wavelength ranges, the number of spectral bands, and degradation mechanisms can severely impact model performance in real-world applications. Third, CNN-based methods continue to suffer from limited interpretability. While numerous approaches have attempted to address these inherent challenges [40], [41], none have comprehensively resolved all issues.

C. Model-Based Methods

Compared to pansharpening-based and CNN-based methods, model-based approaches offer enhanced modeling flexibility and greater interpretability, which makes them a promising alternative in certain contexts. Model-based methods construct relationships between the observed MSI, HSI, and the latent high-resolution HSI from a data generation perspective, known as degradation models or imaging models. Through iterative optimization strategies, the latent HR-HSI is progressively estimated with regularization. Based on differences in spatial and spectral information modeling strategies, model-based methods can be further divided into matrix-based and tensor-based paradigms.

The matrix-based approach decomposes the latent HR-HSI into the product of the spectral basis and coefficients, and estimates these two factors from the observed HR-MSI and LR-HSI data. Some studies [42], [43] employed the spectral basis and coefficients derived from LR-HSI and HR-MSI, respectively, and then constructed relational models containing the spectral and spatial fidelity terms, respectively. Later studies [44], [45] argued that LR-HSI also contains spatial information, and joint modeling of HR-MSI and LR-HSI can lead to more accurate estimation. They used LR-HSI to estimate the spectral basis and then estimated the coefficients using both HR-MSI and LR-HSI. For example, Simoes et al. [46] employed vertex component analysis to estimate the spectral basis, which was then incorporated into an optimization objective function with total variation regularization to estimate the coefficients. Ren et al. [47], considering differences among various targets, applied image segmentation techniques in the fusion problem, and used local optimization to separately estimate the spectral basis and coefficients. Additionally, some studies have directly used known degradation models to establish the relationship between HR-HSI and the observed HR-MSI and LR-HSI without requiring matrix decomposition. The primary focus of these approaches was to enhance estimation accuracy by regularizing the model parameters with regularizers such as sparsity inducing penalty [48], [49], low-rank inducing penalty [50], [51], [52], non-local self-similarity inducing penalty [53], and total variation penalty [46].

Tensor-based approach asserts that the matrix paradigm destroys the native structure of HSI, leading to partial loss of spatial and spectral information. These approaches integrate degradation models with various tensor decomposition methods, such as canonical polyadic (CP) decomposition [54], Tucker decomposition [55], tensor singular value decomposition (t-SVD) [56], [57], tensor train (TT) decomposition [58], and tensor ring (TR) decomposition [59], [60], to better preserve the inherent structure of hyperspectral data. For example, Kanatsoulis et al. [61] reformulated the HSI-MSI fusion problem into a coupled tensor decomposition framework based on CP decomposition. Inspired by this, Prévost et al. [62] assumed that the latent HSI has approximately low multilinear rank and reformulated the fusion problem into a coupled Tucker tensor approximation. Dian et al. [58] proposed a low TT rank regularization term to constrain the fusion results, fully exploiting the non-local spatial–spectral similarity. Additionally, Xu et al. [57] proposed an iterative regularization optimization method based on tensor subspace representation, termed IR-TenSR.

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D. Degradation Model

We now present a review of the degradation model. In this work, an N-dimensional tensor is represented as $\mathcal{M} \in \mathbb{R}^{I_1 \times I_2, \ldots, \times I_N}$. Its mode-$n$ matricization is represented as the matrix $\mathbf{M}_{(n)} \in \mathbb{R}^{I_n \times I_1 I_2, \ldots, I_{n-1} I_{n+1}, \ldots, I_N}$. In the literature, low-spatial-resolution HSIs (LR-HSI), HR-MSI, and HR-HSI are typically represented as 3-D tensors, where the first, second, and third modes correspond to the image width, height, and spectral dimension, respectively. The HR-HSI to be estimated is represented as $\mathcal{Z} \in \mathbb{R}^{N_W \times N_H \times N_S}$, where N_W and N_H denote the spatial dimensions, and N_S denotes the spectral dimension. Correspondingly, the observed LR-HSI and HR-MSI are denoted as $\mathcal{X} \in \mathbb{R}^{N_w \times N_h \times N_S}$ and $\mathcal{Y} \in \mathbb{R}^{N_W \times N_H \times N_s}$, respectively. Note that it typically holds that $N_w < N_W$, $N_h < N_H$ and $N_s < N_S$.

In the degradation model, $\mathcal{X}$ is assumed to be obtained from $\mathcal{Z}$ through spatial downsampling. This modeling equation can be written as

$$\mathbf{X}_{(3)} = \mathbf{Z}_{(3)}\mathbf{Q}\mathbf{S} + \boldsymbol{\varepsilon}_1 \tag{1}$$

where $\boldsymbol{\varepsilon}_1$ represents Gaussian white noise. $\mathbf{X}_{(3)} \in \mathbb{R}^{N_S \times N_w N_h}$ and $\mathbf{Z}_{(3)} \in \mathbb{R}^{N_S \times N_W N_H}$ are the mode-3 matricizations of tensors $\mathcal{X}$ and $\mathcal{Z}$, respectively. The matrix $\mathbf{Q} \in \mathbb{R}^{N_W N_H \times N_W N_H}$ represents the PSF of the hyperspectral imaging sensor in matrix form, also known as the spatial response. It is a block circulant matrix, where each column stands for the spatial point spread effect. The matrix $\mathbf{S} \in \mathbb{R}^{N_W N_H \times N_w N_h}$ denotes the spatial downsampling matrix, which selects the corresponding pixels based on the sampling factor $r = \sqrt{N_W N_H / N_w N_h}$ [50].

The observable HR-MSI can be modeled as being obtained from $\mathcal{Z}$ through spectral downsampling. This relationship can be written as

$$\mathbf{Y}_{(3)} = \mathbf{R}\mathbf{Z}_{(3)} + \boldsymbol{\varepsilon}_2 \tag{2}$$

where $\boldsymbol{\varepsilon}_2$ represents Gaussian white noise, and the matrix $\mathbf{Y}_{(3)} \in \mathbb{R}^{N_s \times N_W N_H}$ is the mode-3 matricization of the tensor $\mathcal{Y}$. Additionally, the matrix $\mathbf{R} \in \mathbb{R}^{N_s \times N_S}$ represents the SRF in matrix form. The SRF models the spectral sensitivity of the sensor, describing how different spectral bands are combined to form the observed measurements. This matrix performs a downsampling operation along the spectral dimension by aggregating the high-spectral-resolution hyperspectral data into the lower-spectral-resolution multispectral format [52].

III. Degradation Parameters Estimation

In this work, we adopt a perspective rooted in FDA to estimate the degradation parameters and recover the high-resolution HSI (HR-HSI). Specifically, we model the true underlying HR-HSI signal as a continuous function, $\mathcal{Z}(i, j, \lambda)$, where i and j represent the spatial indices, and λ denotes the spectral index. Here, i, j, and λ are treated as continuous indices, and the function $\mathcal{Z}(i, j, \lambda)$ is assumed to be locally smooth with respect to these variables. The processes of observing HR-MSI $\mathcal{Y}$ and LR-HSI $\mathcal{X}$, which have been reviewed in Section II-D, can be effectively described using functional representations. FDA inherently respects the smoothness of the data and the model parameters, ensuring that the underlying continuity and local structure of the data are preserved. By leveraging the low intrinsic dimensionality of the hyperspectral data and the degradation parameters, this approach allows for more effective and accurate estimation of the associated model parameters, thereby improving performance in HR-HSI reconstruction.

A. Function View of the Degradation Model in HSI-MSI Fusion

The process of observing HR-MSI $\mathcal{Y}$ is by applying SRF to $\mathcal{Z}(i, j, \lambda)$, and recording the value at several discrete indices. The SRF is treated as a continuous, non-negative, smooth function. We denote $\phi_\beta(\lambda)$ as SRF for multispectral bands $\beta = 1, \ldots, N_s$. The radiance intensity of a single band β in multispectral $\mathcal{Y}$ results from the convolution of the hyperspectral signal $\mathcal{Z}$ with the SRF

$$\mathcal{Y}(i, j, \beta) = \int \mathcal{Z}(i, j, \lambda)\phi_\beta(\lambda)d\lambda + \varepsilon \tag{3}$$

where i, j denote the spatial location indices for $i = 1, \ldots, N_W$ and $j = 1, \ldots, N_H$, and ε represents Gaussian noise. This equation characterizes the relationship between HR-MSI $\mathcal{Y}$ and the latent HR-HSI signal $\mathcal{Z}$ in a function representation.

We can similarly consider the relationship between LR-HSI $\mathcal{X}$ and HR-HSI $\mathcal{Z}$ from the function perspective. The low-resolution $\widetilde{\mathcal{X}}$ is obtained by first convolving the underlying high-resolution $\mathcal{Z}$ with a spatial blur kernel Q

$$\widetilde{\mathcal{X}}(i, j, \lambda) = \int \mathcal{Z}(i - u, j - v, \lambda)Q(u, v)dudv + \varepsilon \tag{4}$$

for $i = 1, \ldots, N_W$ and $j = 1, \ldots, N_H$. In the above, the function Q typically has a compact support in a neighbor around the origin $(u, v) = (0, 0)$. After the convolution, $\widetilde{\mathcal{X}}(i, j, \lambda)$ is spatially downsampled by a factor of r to get the LR-HSI observation $\mathcal{X}$.

B. Proposed Estimation Procedure

We now present a novel approach to precisely estimate the degradation parameters Q and ϕ_β, which are viewed as functions in our framework. Based on the above function models (3) and (4), we can deduce that the following formula holds:

$$\int \mathcal{Y}(i - u, j - v, \beta)Q(u, v)dudv = \int \widetilde{\mathcal{X}}(i, j, \lambda)\phi_\beta(\lambda)d\lambda + \varepsilon \tag{5}$$

where ε represents Gaussian noise. It is important to note that the values on both sides of the above equation can be computed for the spatial indices (i, j) corresponding to the LR-HSI dataset. This equation establishes a connection between the LR-HSI and HR-MSI data by further applying spatial blurring to the HR-MSI data and spectral downsampling to the LR-HSI data. It parallels the formulations commonly expressed in matrix form found in the literature [40]. Our functional framework leverages the smoothness properties of $Q(u, v)$ and $\phi_\beta(\lambda)$, enabling more efficient and accurate parameter estimation.

Based on model (5), we can estimate the functional parameters $Q(u, v)$ and $\phi_\beta(\lambda)$. At a high level, this estimation is approached using an alternating minimization scheme, where

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the following two subproblems are solved iteratively. Given the current estimate $Q^{(k-1)}(u, v)$ of $Q(u, v)$, define

$$H_{i,j,\beta}^{(k)} = \int \mathcal{Y}(i-u, j-v, \beta) Q^{(k-1)}(u, v) du dv \tag{6}$$

and solve

$$\phi_\beta^{(k)}(\lambda) = \arg\min_{\phi_\beta} F(\phi_\beta) \tag{7}$$
$$F(\phi_\beta) = \sum_{i,j} \left[H_{i,j,\beta}^{(k)} - \int \widetilde{\mathcal{X}}(i, j, \lambda) \phi_\beta(\lambda) d\lambda \right]^2.$$

In the above, the summation is taken for the spatial indices (i, j) corresponding to the LR-HSI dataset. Meanwhile, given the current estimate $\phi_\beta^{(k)}$ of ϕ_β, define

$$E_{i,j,\beta}^{(k)} = \int \widetilde{\mathcal{X}}(i, j, \lambda) \phi_\beta^{(k)}(\lambda) d\lambda \tag{8}$$

and solve

$$Q^{(k)}(u, v) = \arg\min_{Q \in \mathcal{Q}} G(Q) \tag{9}$$
$$G(Q) = \sum_{i,j,\beta} \left[E_{i,j,\beta}^{(k)} - \int \mathcal{Y}(i-u, j-v, \beta) Q(u, v) du dv \right]^2$$

where superscript (k) indicates the kth iteration. In the above, the summation is also taken for the spatial indices (i, j) corresponding to the LR-HSI dataset. To obtain more physically meaningful estimation results, the solution in (9) should also satisfy two constraints.

1) *Pointwise summation normalization:*

$$\int Q(u, v) du dv = 1. \tag{10}$$

2) *Non-negativity:*

$$\forall u, v, Q(u, v) \geq 0. \tag{11}$$

In (9), we use $\mathcal{Q}$ to denote the set of functions satisfying the above two constraints.

By modeling the SRF and PSF functions using FDA techniques and iteratively solving (7) and (9), we obtain estimates for the PSF and the SRF of each band. Algorithm 1 summarizes the entire process of estimating the spectral and spatial degradation parameters. In the experiments of this work, the final estimates are obtained after reaching the maximum number of iterations $K = 50$. In Section III-C, we will introduce the implementation details for solving (7) and (9) in the framework of FDA.

C. Implementation Details

For the optimization (7), we rewrite $\widetilde{\mathcal{X}}(i, j, \lambda)$ as $\widetilde{\mathcal{X}}_{i,j}(\lambda)$ and have that

$$\phi_\beta^{(k)}(\lambda) = \arg\min_{\phi_\beta} \sum_{i,j} \left[H_{i,j,\beta}^{(k)} - \int \widetilde{\mathcal{X}}_{i,j}(\lambda) \phi_\beta(\lambda) d\lambda \right]^2. \tag{12}$$

From the above, we can see (12) is a typical functional linear model that has been extensively studied [63], [64]. Therefore, we can use FDA techniques to estimate the SRF ϕ_β. This approach has three main advantages: first, FDA considers the intrinsic correlations between discrete variables through functional modeling, resulting in more stable and accurate estimates. Second, the FDA does not require additional priors or artificial penalty constraints, which enhances the generalization ability of the algorithm. Third, the ideal SRF should be continuous and smooth, which aligns with the fundamental assumptions of FDA techniques.

Algorithm 1 Proposed Method for Estimating Degradation Parameters

Input: LR-HSI $\mathcal{X}$ and HR-MSI $\mathcal{Y}$
Output: Spectral response ϕ and spatial response Q
Initialize: Perform an interpolation upsampling operation on $\mathcal{X}$ with a factor r, resulting in the upsampled $\widetilde{\mathcal{X}}$. Initialize the blur kernel $Q^{(0)}$ as a random matrix;

for $k = 1 : K$ **do**
  Update $\mathbf{H}^{(k)}$ via
  $$H_{i,j,\beta}^{(k)} = \int \mathcal{Y}(i-u, j-v, \beta) Q^{(k-1)}(u, v) du dv;$$
  Update $\phi_\beta^{(k)}$ by solving (7);
  Update $\mathbf{E}^{(k)}$ via
  $$E_{i,j,\beta}^{(k)} = \int \widetilde{\mathcal{X}}(i, j, \lambda) \phi_\beta^{(k)}(\lambda) d\lambda;$$
  Update $Q^{(k)}$ by solving (9);
  $k = k + 1$;
end for

The first step in applying the FDA is to expand the function ϕ_β as a linear combination of basis functions

$$\phi_\beta(\lambda) = \sum_{l=1}^{L} b_{\beta l} \varphi_l(\lambda) \tag{13}$$

where $\varphi_l(\lambda)$'s are a set of spline basis, L is the specified number of basis, and $\boldsymbol{b}_\beta = (b_1, \ldots, b_l)'$ is the coefficient vector of the basis. Using the expansion in (13), the estimation of $\phi_\beta(\lambda)$ translates into the estimation of $\boldsymbol{b}_\beta$. Therefore, the optimization problem (12) can be reformulated as

$$\hat{\boldsymbol{b}}_\beta = \arg\min_{\boldsymbol{b}_\beta} \sum_{i,j} \left[H_{i,j,\beta} - \sum_{l=1}^{L} b_{\beta l} \int \widetilde{\mathcal{X}}_{i,j}(\lambda) \varphi_l(\lambda) d\lambda \right]^2. \tag{14}$$

In the optimization (14), the hyperspectral observation vector and the basis functions are known, making it easy to obtain the estimated result for $\boldsymbol{b}_\beta$. Subsequently, by substituting the estimated coefficient $\boldsymbol{b}_\beta$ into (13), the updated estimation result of the SRF $\phi_\beta(\lambda)$ is obtained.

In all the experiments conducted in this article, we choose cubic B-spline basis as the functional basis [65], which is defined as follows:

$$\varphi_l(\lambda) = B_{l,3}(\lambda) \tag{15}$$

where $B_{l,3}(\lambda)$ represents the cubic B-spline basis function and is recursively defined as

$$B_{l,0}(v) = \begin{cases} 1, & \text{if } t_l \leq v \leq t_{l+1} \\ 0, & \text{otherwise} \end{cases} \tag{16}$$

$$B_{l,3}(\lambda) = \frac{\lambda - t_l}{t_{l+3} - t_l} B_{l,2}(\lambda) + \frac{\lambda - t_l}{t_{l+3} - t_l} B_{l+1,2}(\lambda) \tag{17}$$

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where t_l are the knots that define the intervals for the B-spline basis. The number of functional basis L is predetermined.

For the optimization (9), we rewrite $\mathcal{Y}(i-u, j-v, \beta)$ as $\mathcal{Y}_\beta(i-u, j-v)$, and its objective function can be written as

$$G(Q) = \sum_{i,j,\beta} \left[E^{(k)}_{i,j,\beta} - \int \mathcal{Y}_\beta(i-u, j-v) Q(u,v) du dv \right]^2. \tag{18}$$

The objective function (18) can also be seen as a typical loss for the functional linear model, where the convolution kernel Q simulates the image blurring process. The blurring in remote sensing images is caused by the optical imaging system. PSF is typically represented by a 2-D Gaussian filter [66]. In this work, the convolution kernel is expressed as a linear combination of tensor product splines

$$Q(u,v) = \sum_{m=1}^{M} \sum_{n=1}^{N} c_{mn} \varphi_m(u) \varphi_n(v) \tag{19}$$

where $\varphi_m(u)$ and $\varphi_n(v)$ represent the basis in the horizontal and vertical directions, respectively. In the above, M and N denote the number of basis in each direction, and c_{mn} represents the combination coefficient. We put the coefficient into a matrix $\mathbf{C} = (c_{mn})$ of size $M \times N$.

Given the basis expansion (19), the two constraints (10) and (11) can be equivalently formulated. The first constraint can be simplified to

$$\int \sum_{m=1}^{M} \sum_{n=1}^{N} c_{mn} \varphi_m(u) \varphi_n(v) du dv = 1. \tag{20}$$

Due to the non-negative property of B-spline basis, the second constraint (11) is equivalent to imposing $c_{mn} \geq 0$ for all m, n. The constraint set $\mathcal{Q}$ in (9) can be represented for the matrix $\mathbb{C}$ as

$$\mathcal{C} = \Big\{ \mathbf{C} = (c_{mn}) : c_{mn} \geq 0 \text{ and } \int \sum_{m=1}^{M} \sum_{n=1}^{N} c_{mn} \varphi_m(u) \varphi_n(v) du dv = 1 \Big\}. \tag{21}$$

By substituting the blur kernel represented by these basis functions (19) into optimization (9), we obtain

$$\widehat{\mathbf{C}} = \arg\min_{\mathbf{C} \in \mathcal{C}} G(\mathbf{C}) \tag{22}$$

$$G(\mathbf{C}) = \sum_{i,j,\beta} \left[E_{i,j,\beta} - \sum_{m,n} c_{mn} \bar{Y}^{m,n}_{i,j,\beta} \right]^2$$

$$\text{and } \bar{Y}^{m,n}_{i,j,\beta} = \int \mathcal{Y}_\beta(i-u, j-v) \varphi_m(u) \varphi_n(v) du dv.$$

The intermediate variable $E_{i,j,\beta}$, the multispectral image $\mathcal{Y}$, and the basis functions are known. Therefore, the coefficients matrix $\widehat{\mathbf{C}} = (\hat{c}_{mn})$ can be obtained by solving the optimization (22). Finally, substitute the estimated $\widehat{\mathbf{C}}$ into formula (19) to obtain the updated estimate of the blur kernel Q.

IV. Fusion Algorithm

A. Spatial Content-Aware Spectral Subspace Regularizer

With the estimated degradation parameters, the next step is to recover the signal function $\mathcal{Z}(i, j, \lambda)$. In this work, we estimate the spectral function for each high-resolution index $i = 1, \ldots, N_W$ and $j = 1, \ldots, N_H$. For convenience, we denote $\mathcal{Z}(i, j, \lambda)$ as $\mathcal{Z}_{i,j}(\lambda)$ in this section. As introduced in Section I, our goal is to develop a spatial content-aware regularizer for their effective recovery.

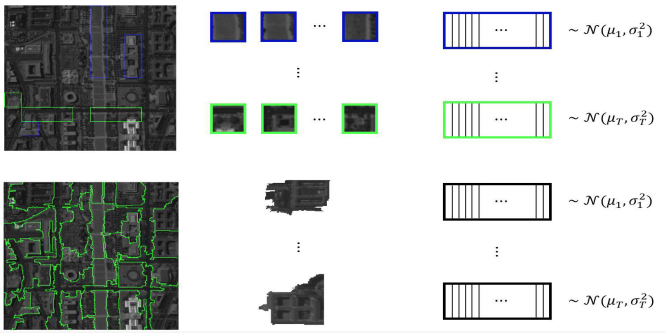


Fig. 1. Block partitioning and spatial content partitioning.

Based on spatial content, we partition the data into spectrally homogeneous subgroups and promote a low-dimensional functional subspace within each group. In particular, for $\mathcal{Z}_{i,j}(\lambda)$ with $i = 1, \ldots, N_W$ and $j = 1, \ldots, N_H$, we treat them as functions originating from a mixture distribution with T components. Each mixture component corresponds to a distinct type of spatial content. Therefore, the indices (i, j) are partitioned into T groups (or superpixel), denoted as $\mathcal{T}_1, \ldots, \mathcal{T}_T$. We employ a superpixel method [23] on MSI to obtain these partitioning blocks. Fig. 1 (bottom left) illustrates the result of such spatial partitioning. The hyperspectral data within each group can be well approximated by a low-dimensional subspace. Specifically, for (i, j) belonging to the tth group (i.e., $(i, j) \in \mathcal{T}_t$), the data can be expressed as

$$\mathcal{Z}_{i,j}(\lambda) \approx \mu_t(\lambda) + \sum_{r=1}^{R^t} \alpha_{i,j} \psi_{t,r}(\lambda). \tag{23}$$

Equation (23) states that each spectral function $\mathcal{Z}_{i,j}(\lambda)$ can be approximated by a linear combination of the mean function $\mu_t(\lambda)$ and several principal component functions $\psi_{t,r}(\lambda)$. We refer to the group-wise low rank assumption in (23) as the spatial content-aware spectral subspace assumption.

Employing our assumption (23) generally has stronger regularization effect for denoising and recovering $\mathcal{Z}_{i,j}(\lambda)$. To illustrate this, we fit the model (23) to the partitioned hyperspectral data in Fig. 1. Assuming $\alpha_{i,j}$ follows a Gaussian distribution, we compute the entropy of the probability distribution of the low-rank model for each partition. Entropy serves as a measure of the uncertainty in a distribution, where smaller entropy values indicate a stronger and more informative model, particularly when it acts as a prior for regularization. For comparison, we also compute the entropy for a global low-rank model imposed without spatial partitioning, denoted as "matrix." Additionally, we evaluate the entropy for a model where spatial pixels are partitioned in a rectangular manner and then clustered, denoted as "block." Fig. 2 shows the boxplot of the entropy values of every partitions for distinct choices of model rank R. Our model (which is based on the "superpixel" method) generally has smaller entropy values. The comparison in Fig. 2 highlights the effectiveness

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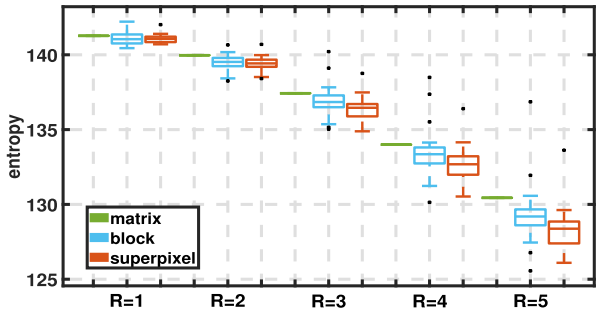


Fig. 2. Entropy of the distribution of various partitioning methods across different model ranks R. Our model is based on the "superpixel" partitioning.

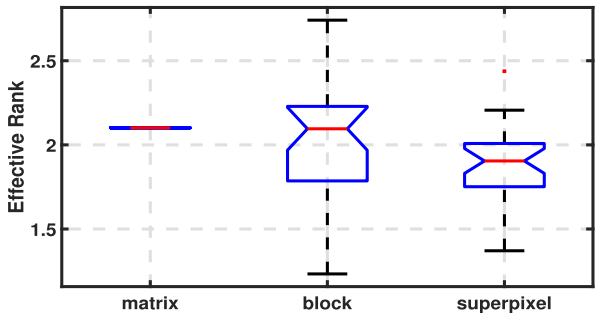


Fig. 3. Effective rank of different partitioning methods. Our model is based on the "superpixel" partitioning.

of our approach in leveraging spatially partitioned low-rank assumptions for enhanced regularization.

In addition, as shown in Fig. 3, we compare the effective ranks of different partitioning methods. The concept of effective rank, originally introduced by Roy and Vetterli [67], measures the effective dimensionality of a matrix based on its singular value distribution. In this work, the effective rank is calculated as the ratio of the sum of singular values to the largest singular value. A smaller value indicates greater homogeneity within each partition. Our model, based on the "superpixel" partitioning method, demonstrates lower effective ranks for the mixture components, as shown in the boxplots in Fig. 3. This indicates that our approach can better constrain model parameters within a compact and informative subspace, thereby enhancing the low-rank characteristics of the spatial regions. This enhances the model's performance in estimation and regularization, leading to more effective processing of hyperspectral data.

Moreover, most existing methods in the literature utilize spatial and spectral regularization independently, neglecting their interdependence. These methods often impose constraints directly in the spectral domain without fully leveraging the smoothness property inherent to each $\mathcal{Z}_{i,j}(\lambda)$. This smoothness property implies that, although $\mathcal{Z}_{i,j}(\lambda)$ is digitally recorded as a high-dimensional vector, its intrinsic dimension is relatively lower. Properly exploiting this property can reduce the number of parameters and improve estimation efficiency. For this purpose, we expand the spectral vector using basis functions

$$\mathcal{Z}_{i,j}(\lambda) = \psi^T(\lambda) b_{i,j} \tag{24}$$

where i, j denotes the spatial location indices, ψ represents the basis vectors of dimension L, and $b_{i,j}$ is the basis coefficient vector at location (i, j). For hyperspectral observations in the tth group, we put them into the columns of a matrix, denoted as $\mathbf{Z}^t_{(3)}$. We put their corresponding spline coefficients into another matrix $\bar{\mathbf{B}}^t_{(3)}$. The two matrices are related via

$$\mathbf{Z}^t_{(3)} = \mathbf{\Phi} \bar{\mathbf{B}}^t_{(3)}. \tag{25}$$

In the above, $\mathbf{\Phi}$ represents the basis function matrix. Each row corresponds to an evaluation of the basis functions at one specific recorded λ. By expanding the original discrete spectral vectors into a basis function representation, the inherent structure of the spectral information can be effectively extracted, preserving the essential characteristics of the spectrum. This approach reduces the influence of noise, resulting in more stable spectral information estimation.

To promote low rankness within each super-pixel group, we define the spatial content-aware SSR

$$\mathcal{P}(\mathcal{Z}) = \sum_{t=1}^{T} \left\| \bar{\mathbf{B}}^t_{(3)} \right\|_* \tag{26}$$

where $\|\cdot\|_*$ denotes the nuclear norm, which is defined as the sum of the singular values of a matrix. The proposed regularizer encourages a low-rank basis coefficient matrix for each superpixel of similar spatial content. This is equivalent to encouraging a low-dimensional functional subspace within each group [see also (23)].

B. Proposed FDAssr for HSI-MSI Fusion

Recall our target is to estimate the spectral function for each high-resolution index $i = 1, \ldots, N_W$ and $j = 1, \ldots, N_H$. By integrating the accurate spatial and spectral response parameters estimated in Section III with the efficient spatial content-aware SSR in Section IV-A, we propose the following fusion model for estimating $\mathcal{Z}$:

$$\begin{aligned} \underset{\mathcal{Z}}{\operatorname{argmin}} & \sum_{i,j,\lambda} \left[\widetilde{\mathcal{X}}(i,j,\lambda) - \int \mathcal{Z}(i-u, j-v, \lambda) Q(u,v) du dv \right]^2 \\ & + \sum_{\beta,i,j} \left[\mathcal{Y}(i,j,\beta) - \int \mathcal{Z}(i,j,\lambda) \phi_\beta(\lambda) d\lambda \right]^2 + \mu \mathcal{P}(\mathcal{Z}) \end{aligned} \tag{27}$$

where $\mathcal{P}(\mathcal{Z})$ denotes the regularization term (26), and μ is the tuning parameter. In the above, the summation is taken with respect to all high-resolution indices $i = 1, \ldots, N_W$ and $j = 1, \ldots, N_H$. Our method (27) to recover HR-HSI data via FDA framework and the proposed regularization (26) is termed functional data analysis with spatial content-aware spectral subspace regularizer (FDAssr). We expand each spectral at spatial index (i, j) by basis function as in (24), and re-express the optimization (27) as

$$\begin{aligned} \underset{\bar{\mathbf{B}}}{\operatorname{argmin}} & \sum_{i,j,\lambda} \left[\widetilde{\mathcal{X}}_{i,j,\lambda} - \sum_{u,v} \psi^T(\lambda) b_{i-u,j-v} Q(u,v) \right]^2 \\ & + \sum_{\beta,i,j} \left[\mathcal{Y}_{i,j,\beta} - \sum_{\lambda} \psi^T(\lambda) b_{i,j} \phi_\beta(\lambda) \Delta_\lambda \right]^2 + \mu \sum_{t=1}^{T} \left\| \bar{\mathbf{B}}^t_{(3)} \right\|_*. \end{aligned} \tag{28}$$

In the above, $\bar{\mathbf{B}} \in \mathbb{R}^{N_W \times N_H \times L}$ is a tensor containing all spline coefficients in (24) for all high resolution spatial indices $i = 1, \ldots, N_W$ and $j = 1, \ldots, N_H$. Comparing the first lines

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of (27) and (28), the integral with respect to u, v is replaced by the summation as an approximation. This is because we only aim to recover the spectral at the high-resolution spatial indices, and summation reduces the overhead of numerical integration for faster optimization iterations. Similarly, in the second lines of (27) and (28), the integral with respect to λ is also replaced by summation as an approximation, and Δ_λ is the spacing between λ's. In matrix format, the optimization problem (28) is similar to traditional maximum a posteriori (MAP) estimation models (e.g., [68]). A key difference is that our matrix values are evaluated based on spline basis functions, and we employ our spatial content-aware SSR for effective estimation.

C. Implementation Details

The optimization (28) can be reformulated in matrix notations for the convenience of designing optimization algorithm. In fact, (28) is equivalent to the following:

$$\underset{\bar{\mathbf{B}}_{(3)}}{\operatorname{argmin}} \left\|\mathbf{X}_{(3)} - \boldsymbol{\Phi}\bar{\mathbf{B}}_{(3)}\mathbf{QS}\right\|_F^2 + \left\|\mathbf{Y}_{(3)} - \mathbf{R}\boldsymbol{\Phi}\bar{\mathbf{B}}_{(3)}\right\|_F^2 + \mu\sum_{t=1}^{T}\left\|\bar{\mathbf{B}}_{(3)}^t\right\|_*. \tag{29}$$

In the above, the matrix $\mathbf{Q}$ represents the spatial blur process, its elements are the values of the function $Q(u, v)$ evaluated at appropriate u, v. Meanwhile, $\mathbf{S}$ represents the spatial downsampling process with a sampling rate of r. The SRFs $[\phi_\beta(\lambda)]$ are used to construct the rows of the spectral response matrix $\mathbf{R}$.

The optimization problem (29) is an optimization problem with nuclear norm penalties, and we use the alternating direction method of multipliers (ADMM) framework to efficiently solve it. By introducing the auxiliary variable $\mathbf{Z}_{(3)} = \boldsymbol{\Phi}\bar{\mathbf{B}}_{(3)}, \mathbf{W} = \bar{\mathbf{B}}_{(3)}$, we obtain the following augmented Lagrangian function:

$$\begin{aligned} &L\left(\bar{\mathbf{B}}_{(3)}, \mathbf{Z}_{(3)}, \mathbf{W}, \mathbf{G}_1, \mathbf{G}_2\right) \\ &= \left\|\mathbf{X}_{(3)} - \mathbf{Z}_{(3)}\mathbf{QS}\right\|_F^2 + \left\|\mathbf{Y}_{(3)} - \mathbf{RZ}_{(3)}\right\|_F^2 + \mu\sum_{t=1}^{T}\left\|\mathbf{W}^t\right\|_* \\ &+ \frac{\rho}{2}\left\|\mathbf{Z}_{(3)} - \boldsymbol{\Phi}\bar{\mathbf{B}}_{(3)} + \frac{\mathbf{G}_1}{\rho}\right\|_F^2 + \frac{\rho}{2}\left\|\mathbf{W} - \bar{\mathbf{B}}_{(3)} + \frac{\mathbf{G}_2}{\rho}\right\|_F^2 \end{aligned} \tag{30}$$

where $\mathbf{G}_1$ and $\mathbf{G}_2$ are the Lagrange multipliers, and ρ is the penalty parameter. We solve (30) by iteratively solving the following subproblems.

1) Update $\mathbf{Z}_{(3)}$: Note given the other optimization variables, $\mathbf{Z}_{(3)}$ can be updated by

$$\begin{aligned} \mathbf{Z}_{(3)} &\in \underset{\mathbf{Z}_{(3)}}{\operatorname{argmin}}\, L\left(\bar{\mathbf{B}}_{(3)}, \mathbf{Z}_{(3)}, \mathbf{W}, \mathbf{G}_1, \mathbf{G}_2\right) \\ &= \underset{\mathbf{Z}_{(3)}}{\operatorname{argmin}} \left\|\mathbf{X}_{(3)} - \mathbf{Z}_{(3)}\mathbf{QS}\right\|_F^2 + \left\|\mathbf{Y}_{(3)} - \mathbf{RZ}_{(3)}\right\|_F^2 \\ &+ \frac{\rho}{2}\left\|\mathbf{Z}_{(3)} - \boldsymbol{\Phi}\bar{\mathbf{B}}_{(3)} + \frac{\mathbf{G}_1}{\rho}\right\|_F^2. \end{aligned} \tag{31}$$

The above optimization is strongly convex, so we directly set the derivative with respect to $\mathbf{Z}_{(3)}$ to zero, obtaining the following formula:

$$\mathbf{Z}_{(3)}\mathbf{H}_1 + \mathbf{H}_2\mathbf{Z}_{(3)} = \mathbf{H}_3 \tag{32}$$

where

$$\mathbf{H}_1 = \mathbf{QS}(\mathbf{QS})^T \tag{33}$$

$$\mathbf{H}_2 = \mathbf{R}^T\mathbf{R} + \frac{\rho}{2}\mathbf{I} \tag{34}$$

$$\mathbf{H}_3 = \mathbf{X}_{(3)}(\mathbf{QS})^T + \mathbf{R}^T\mathbf{Y}_{(3)} + \frac{\rho}{2}\boldsymbol{\Phi}\bar{\mathbf{B}}_{(3)} - \frac{1}{2}\mathbf{G}_1 \tag{35}$$

where $\mathbf{I} \in \mathbb{R}^{N_S \times N_S}$ is the identity matrix. Equation (32) is a classic Sylvester equation [69], and its solution is unique if and only if $\mathbf{H}_1$ and $-\mathbf{H}_2$ have no common eigenvalues. From (33) and (34), we know that $\mathbf{H}_1$ is a semi-positive matrix and $\mathbf{H}_2$ is positive definite, meaning the eigenvalues of $-\mathbf{H}_2$ are non-positive. Therefore, the condition for a unique solution is satisfied. Utilizing the properties of $\mathbf{Q}$ and $\mathbf{S}$, we apply a fast solving method [70] to solve (32).

2) Update $\bar{\mathbf{B}}_{(3)}$: Given the other optimization variables, $\bar{\mathbf{B}}_{(3)}$ can be updated by

$$\begin{aligned} \bar{\mathbf{B}}_{(3)} &\in \underset{\bar{\mathbf{B}}_{(3)}}{\operatorname{argmin}}\, L\left(\bar{\mathbf{B}}_{(3)}, \mathbf{Z}_{(3)}, \mathbf{W}, \mathbf{G}_1, \mathbf{G}_2\right) \\ &= \underset{\bar{\mathbf{B}}_{(3)}}{\operatorname{argmin}} \frac{\rho}{2}\left\|\mathbf{Z}_{(3)} - \boldsymbol{\Phi}\bar{\mathbf{B}}_{(3)} + \frac{\mathbf{G}_1}{\rho}\right\|_F^2 \\ &+ \frac{\rho}{2}\left\|\mathbf{W} - \bar{\mathbf{B}}_{(3)} + \frac{\mathbf{G}_2}{\rho}\right\|_F^2. \end{aligned} \tag{36}$$

Similar to the optimization in (31), we set the derivative of (36) with respect to $\bar{\mathbf{B}}_{(3)}$ to zero, resulting in the following equation:

$$\left(\rho\boldsymbol{\Phi}^T\boldsymbol{\Phi} + \rho\mathbf{I}\right)\bar{\mathbf{B}}_{(3)} = \rho\boldsymbol{\Phi}^T\mathbf{Z}_{(3)} + \boldsymbol{\Phi}^T\mathbf{G}_1 + \rho\mathbf{W} + \mathbf{G}_2. \tag{37}$$

Based on the above equation, the updated $\bar{\mathbf{B}}_{(3)}$ can be computed by

$$\bar{\mathbf{B}}_{(3)} = \left(\rho\boldsymbol{\Phi}^T\boldsymbol{\Phi} + \rho\mathbf{I}\right)^{-1}\left(\rho\boldsymbol{\Phi}^T\mathbf{Z}_{(3)} + \boldsymbol{\Phi}^T\mathbf{G}_1 + \rho\mathbf{W} + \mathbf{G}_2\right). \tag{38}$$

3) Update $\mathbf{W}$: Given the other optimization variables, $\mathbf{W}$ can be updated by

$$\begin{aligned} \mathbf{W} &\in \underset{\mathbf{W}}{\operatorname{argmin}}\, L\left(\bar{\mathbf{B}}_{(3)}, \mathbf{Z}_{(3)}, \mathbf{W}, \mathbf{G}_1, \mathbf{G}_2\right) \\ &= \underset{\mathbf{W}}{\operatorname{argmin}}\, \mu\sum_{t=1}^{T}\left\|\mathbf{W}^t\right\|_* + \frac{\rho}{2}\left\|\mathbf{W} - \bar{\mathbf{B}}_{(3)} + \frac{\mathbf{G}_2}{\rho}\right\|_F^2. \end{aligned} \tag{39}$$

Because different superpixel blocks do not overlap, we can optimize each superpixel block separately, resulting in the following formula:

$$\begin{aligned} \mathbf{W}^t &\in \underset{\mathbf{W}^t}{\operatorname{argmin}}\, L\left(\bar{\mathbf{B}}_{(3)}, \mathbf{Z}_{(3)}, \mathbf{W}, \mathbf{G}_1, \mathbf{G}_2\right) \\ &= \underset{\mathbf{W}^t}{\operatorname{argmin}}\, \mu\left\|\mathbf{W}^t\right\|_* + \frac{\rho}{2}\left\|\mathbf{W}^t - \bar{\mathbf{B}}_{(3)}^t + \frac{\mathbf{G}_2^t}{\rho}\right\|_F^2 \end{aligned} \tag{40}$$

where $\bar{\mathbf{B}}_{(3)}^t$ and $\mathbf{G}_2^t$ represent the elements corresponding to the tth superpixel position in $\bar{\mathbf{B}}_{(3)}$ and $\mathbf{G}_2$, respectively. The closed-form solution for (40) is given by

$$\mathbf{W}^t = \mathbf{U}\left(\boldsymbol{\Sigma} - \frac{\mu}{\rho}\right)_+\mathbf{V}^T \tag{41}$$

where $U\Sigma V^T$ is the SVD decomposition of $\bar{\mathbf{B}}_{(3)}^t - \mathbf{G}_2^t/\rho$, and $(\cdot)_+$ denotes taking the positive part. Finally, the optimization results of each superpixel $\mathbf{W}^t$ are combined to obtain the final $\mathbf{W}$.

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4) Update $\mathbf{G}_1$ and $\mathbf{G}_2$: As in the literature of ADMM, the Lagrangian multiplier is updated by

$$\mathbf{G}_1 = \mathbf{G}_1 + \rho\left(\mathbf{Z}_{(3)} - \mathbf{\Phi}\bar{\mathbf{B}}_{(3)}\right) \tag{42}$$

$$\mathbf{G}_2 = \mathbf{G}_2 + \rho\left(\mathbf{W} - \bar{\mathbf{B}}_{(3)}\right). \tag{43}$$

At this point, an iterative optimization cycle is completed. The iteration terminates when the following condition is met:

$$\frac{\left\|\mathbf{Z}_{(3)}^{k+1} - \mathbf{Z}_{(3)}^{k}\right\|_F}{\|\mathbf{Z}_{(3)}^{k}\|_F} < \zeta \tag{44}$$

where k is the number of iterations and ζ is the convergence threshold, set to 0.0001 in this article. The details of the entire optimization process are shown in Algorithm 2.

Algorithm 2 FDAssr Algorithm for HSI-MSI Fusion

Input: Mode-3 unfolded LR-HSI $\mathbf{X}_{(3)}$ and HR-MSI $\mathbf{Y}_{(3)}$
Output: Mode-3 unfolded HR-HSI $\mathbf{Z}_{(3)}$
Preprocessing: Estimate the spectral response $\mathbf{R}$ and spatial response $\mathbf{Q}$ of the matrix norm using Algorithm 1;

1: **while** $\frac{\left\|\mathbf{Z}_{(3)}^{k+1} - \mathbf{Z}_{(3)}^{k}\right\|_2}{\mathbf{Z}_{(3)}^{k}} > \zeta$ **do**
2: Update $\mathbf{Z}_{(3)}$ by solving (31);
3: Update $\bar{\mathbf{B}}_{(3)}$ by solving (36);
4: Update $\mathbf{W}$ by solving (39);
5: Update $\mathbf{G}_1 = \mathbf{G}_1 + \rho(\mathbf{Z}_{(3)} - \mathbf{\Phi}\bar{\mathbf{B}}_{(3)})$;
6: Update $\mathbf{G}_2 = \mathbf{G}_2 + \rho(\mathbf{W} - \bar{\mathbf{B}}_{(3)})$;
7: **end while**

V. Experiments

To validate the effectiveness of our proposed method, we conducted a comparative analysis on publicly available satellite remote sensing datasets.

A. Datasets

1) Pavia University: The HSIs of Pavia University [71] were obtained by the reflective optics system imaging spectrometer-3 (ROSIS-3) optical airborne sensor over the area of Pavia University in Italy. The original dimensions are $610 \times 340 \times 115$ with a spatial resolution of 1.3 m. We removed the bands severely affected by noise, retaining 93 bands for the experiment. The MSI was obtained by spectrally subsampling the reference image with the IKONOS-like spectral response [51]. The HSI was obtained by applying a Gaussian filter with a size of 7×7 and a standard deviation of 3 to the reference image, followed by spatial downsampling with a factor of 4.

2) Chikusei: The Chikusei dataset [72] was acquired by the HeadWalls Hyperspec Visible and Near-Infrared imaging sensor over Chikusei, Ibaraki, Japan. The dimensions are $2517 \times 2355 \times 128$ with a spatial resolution of 2.5 m. After removing noisy bands, we extracted a subimage with dimensions of $400 \times 400 \times 110$ for the experiments. The MSI was generated by applying the multispectral SRF of the WorldView-2 satellite to the reference image. The HSI was generated in the same manner as for the Pavia University dataset.

3) Houston: The Houston dataset [73] was acquired by a hyperspectral airborne sensor over The University of Houston in the United States. The dimensions are $349 \times 1905 \times 144$ with a spatial resolution of 2.5 m. We selected a subimage with dimensions of 336×800 for the experiments. The MSI was obtained by averaging the bands of the reference image according to the SRF of the WorldView-2 satellite. The HSI was generated in the same manner as for the Pavia University dataset.

4) Washington DC Mall: The Washington DC Mall dataset [72] was acquired by the hyperspectral digital imagery collection experiment (HYDICE) sensor. The dimensions are $1280 \times 307 \times 191$. We selected a subimage with dimensions of $300 \times 300 \times 100$ for the experiments. The MSI was obtained by averaging the bands of the reference image according to the SRF of the IKONOS satellite. The HSI was generated in the same manner as for the Pavia University dataset.

B. Comparison Methods

In this study, we conducted a comparative analysis of the proposed method (FDAssr) against ten advanced methods. These methods[1] include Gram-Schmidt adaptive (GSA) [74], coupled nonnegative matrix factorization (CNMF) [52], HySure [46], FUSE [70], coupled sparse tensor factorization (CSTF) [75], low tensor multi-rank (LTMR) [56], STEREO [61], BSCOTT [62], IR-TenSR [57], and low tensor-train rank (LTTR) [58]. We chose to compare with traditional and optimization-based methods, excluding CNN-based methods that require training.

C. Quality Metrics

We followed Wald's protocol [76] in our experiments to generate simulated data. Specifically, the originally acquired HSI is used as the ideal HR-HSI reference image, which is then spatially and spectrally downsampled to obtain the corresponding LR-HSI and HR-MSI, respectively. The HSI-MSI fusion methods are then applied to obtain the fusion results, which are evaluated using various quantitative metrics. In this study, we employed seven quantitative metrics to assess the fusion performance in terms of spatial, spectral, and overall quality. These metrics include root mean square error (RMSE) [75], peak signal-to-noise ratio (PSNR), spectral angle mapper (SAM) [77], relative dimensionless global error in synthesis (ERGAS) [78], extended universal image quality index (Q2n) [79], universal image quality index (UIQI) [80], and runtime (seconds).

D. Degradation Parameter Estimation Experiments

In this section, we evaluate the performance of the proposed method (denoted as FDA) in estimating degradation parameters using simulated data. We first generate simulated data based on the Pavia University and Chikusei datasets by applying known spectral and spatial degradation parameters. Then, we estimate the parameters using our proposed method and compare the estimated results with the parameters used for data generation to validate the effectiveness of the estimation approach. For comparison, we also implement the estimation method proposed by Dian et al. [40], which achieves stable

[1]We thank Yokoya et al. [72] for providing the code toolbox that includes most of the methods.

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TABLE I

PSNR VALUES OF SPECTRAL AND SPATIAL RESPONSES ESTIMATED BY THE ZSL [40] AND OUR FDA METHOD UNDER VARIOUS NOISE LEVELS

	Zero-shot		FDA	
	R	**B**	**R**	**B**
PaviaU				
0dB	64.3534	126.1458	135.8528	166.5951
50dB	45.7204	71.5725	54.6017	79.3067
40dB	42.8592	60.9722	49.6153	69.4778
30dB	41.2158	51.2395	45.4082	59.3484
Chikusei				
0dB	46.6358	107.1388	138.4011	170.6789
50dB	40.5188	57.4079	46.2496	78.8602
40dB	37.7293	54.1651	41.5963	66.6145
30dB	35.3875	51.9581	37.5016	59.0364

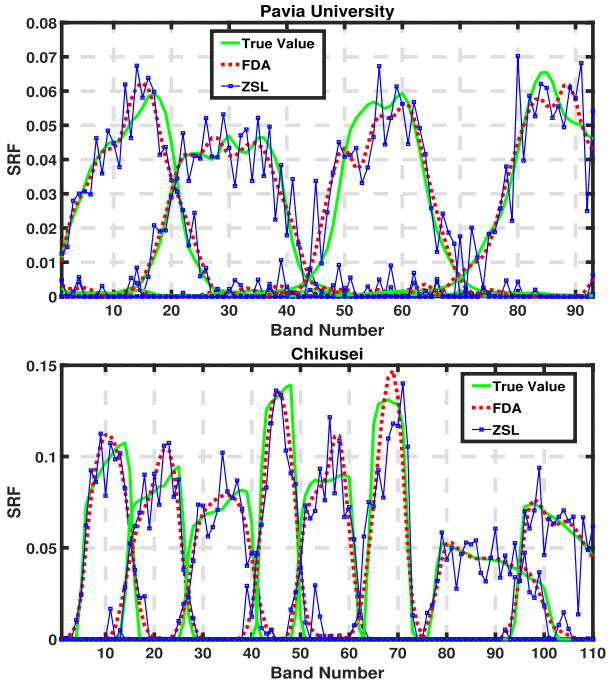


Fig. 4. Simulated SRF and the corresponding SRF curves estimated by the ZSL [40] and our FDA method.

estimates by leveraging a least-squares method with physical prior constraints.

Table I presents the PSNR values of the estimation results. Our method (FDA) achieves consistently high PSNR values across HSI and MSI data subjected to varying levels of Gaussian noise, demonstrating the robustness and effectiveness of the proposed approach. Additionally, Fig. 4 provides a visual comparison of the true SRF and the SRF curves estimated by both methods under 50 dB noise. As illustrated, the SRF estimated by our method exhibits smoother and more stable characteristics, aligning well with the physical properties of an ideal SRF. This further confirms the superiority of the proposed estimation method.

E. Experiments on Simulated Data Fusion

1) Quantitative Comparison: The quantitative evaluation results for the four datasets, Pavia University, Chikusei, Houston, and Washington DC Mall, are presented in Tables II–V, respectively. Our fusion method (FDAssr) achieves the best performance across most quantitative metrics on various datasets. Notably, the CSTF, LTMR, and LTTR methods exhibit strong results on different datasets due to their unique matrix or tensor modeling approaches and prior constraints. Additionally, the HySure method leverages a Bayesian modeling perspective and relies on its accurate endmember-abundance decomposition model, demonstrating clear advantages. Methods such as STEREO and BSCOTT perform well on individual datasets; however, their performance is sensitive the selection of tensor rank, leading to unstable results. Furthermore, Fig. 5 illustrates the PSNR values for each band across the four datasets, demonstrating that our method consistently outperforms the alternatives, which aligns closely with the quantitative findings. Overall, FDAssr surpasses all comparison methods in these simulated tests, providing superior and more stable performance.

TABLE II

OVERALL PERFORMANCE OF METRICS RMSE, PSNR, SAM, ERGAS, Q2N, AND UIQI BY DIFFERENT METHODS ON PAVIA UNIVERSITY DATASET

Method	**RMSE**	**PSNR**	**SAM**	**ERGAS**	**Q2n**	**UIQI**	**Runtimes(s)**
GSA	0.0149	37.1754	3.9984	2.5713	0.8085	0.9617	1.5181
CNMF	0.0193	34.3908	3.3749	3.5225	0.7722	0.9538	28.1965
HySure	0.0075	44.8251	2.4586	1.4742	0.9307	0.9307	914.5517
FUSE	0.0173	35.5889	3.8013	2.9227	0.8024	0.9647	1.1301
CSTF	0.0072	45.0262	2.3612	1.4414	0.9344	0.9837	180.3316
LTMR	0.0087	44.1287	2.7232	1.5834	0.9218	0.9811	204.7699
STEREO	0.0108	41.0614	3.4802	2.0273	0.8777	0.9728	890.0114
BSCOTT	0.0098	42.2612	3.1281	1.9705	0.8957	0.9769	**1.0460**
IR-TenSR	0.0090	43.3152	2.8103	1.6509	0.9139	0.9808	517.8159
LTTR	0.0081	44.2113	2.5800	1.5735	0.9256	0.9815	657.8738
FDAssr	**0.0067**	**46.0234**	**2.1985**	**1.3468**	**0.9387**	**0.9851**	234.0975
Ideal value	0	$+\infty$	**0**	**0**	**1**	**1**	**0**

TABLE III

OVERALL PERFORMANCE OF METRICS RMSE, PSNR, SAM, ERGAS, Q2N, AND UIQI BY DIFFERENT METHODS ON CHIKUSEI DATASET

Method	**RMSE**	**PSNR**	**SAM**	**ERGAS**	**Q2n**	**UIQI**	**Runtimes(s)**
GSA	0.0042	48.5364	0.8239	1.2862	0.8375	0.9861	1.5157
CNMF	0.0080	42.5209	0.8587	2.1655	0.6795	0.9779	24.3578
HySure	0.0026	56.2722	0.5639	0.9805	0.9066	0.9906	538.4900
FUSE	0.0080	42.9034	0.8964	2.0463	0.6463	0.9817	1.6754
CSTF	0.0024	58.5361	0.5271	0.7142	0.9109	0.9906	160.6994
LTMR	0.0031	56.0718	0.6261	0.9445	0.8911	0.9915	143.0238
STEREO	0.0034	52.6533	0.7465	1.0896	0.8665	0.9860	704.7201
BSCOTT	0.0065	52.2793	0.9162	1.6426	0.8529	0.9826	**1.0527**
IR-TenSR	0.0055	49.4967	1.0575	1.2605	0.8471	0.9845	270.7533
LTTR	0.0027	56.2043	0.6020	0.7130	0.8958	0.9905	605.35018
FDAssr	**0.0021**	**59.3623**	**0.4734**	**0.5766**	**0.9175**	**0.9954**	197.5027
Ideal value	0	$+\infty$	**0**	**0**	**1**	**1**	**0**

TABLE IV

OVERALL PERFORMANCE OF METRICS RMSE, PSNR, SAM, ERGAS, Q2N, AND UIQI BY DIFFERENT METHODS ON HOUSTON DATASET

Method	**RMSE**	**PSNR**	**SAM**	**ERGAS**	**Q2n**	**UIQI**	**Runtimes(s)**
GSA	0.0047	46.4637	2.1594	1.3871	0.9661	0.9859	2.9988
CNMF	0.0071	42.6178	2.6502	1.9438	0.9171	0.9773	42.6484
HySure	0.0021	53.6057	0.9480	0.7902	0.9869	0.9944	617.4534
FUSE	0.0077	41.8045	2.2161	2.0279	0.9326	0.9794	2.6718
CSTF	0.0023	52.1680	1.0962	0.7887	0.9927	0.9938	82.2531
LTMR	0.0021	53.8766	0.8392	0.8637	**0.9932**	0.9937	201.2323
STEREO	0.0030	50.9577	1.5251	1.0949	0.9907	0.9878	915.4490
BSCOTT	0.0023	52.4065	0.8976	0.7870	0.9885	0.9941	**1.2805**
IR-TenSR	0.0057	50.5793	1.0978	1.1147	0.8219	0.9851	511.00025
LTTR	0.0036	50.5425	1.7690	1.2350	0.9885	0.9867	1079.3612
FDAssr	**0.0015**	**56.7172**	**0.6098**	**0.5723**	0.9903	**0.9967**	500.2550
Ideal value	0	$+\infty$	**0**	**0**	**1**	**1**	**0**

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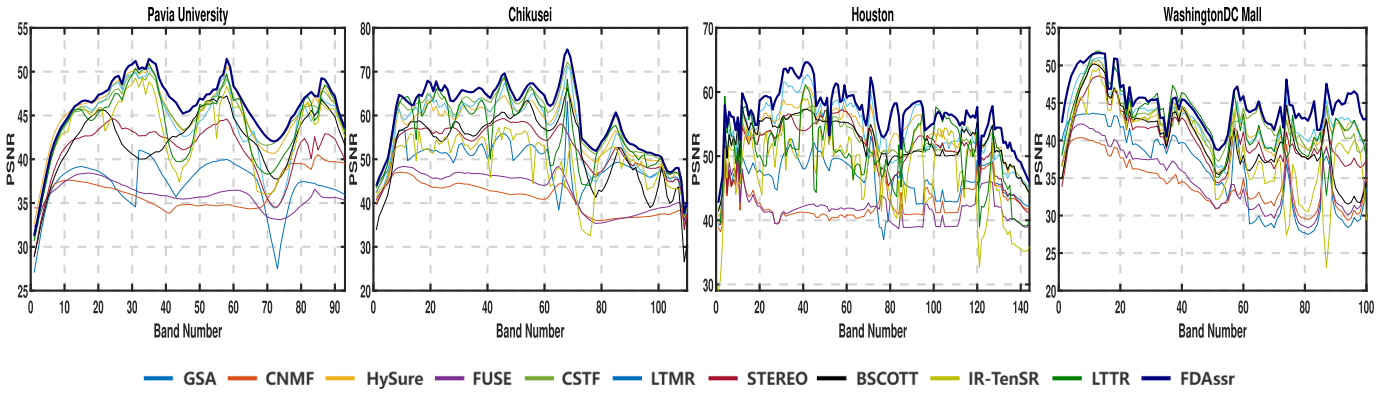


Fig. 5. PSNR values along the spectral direction obtained by the test methods on four different datasets.

TABLE V
OVERALL PERFORMANCE OF METRICS RMSE, PSNR, SAM, ERGAS, Q2N, AND UIQI BY DIFFERENT METHODS ON WASHINGTON DC MALL DATASET

Method	RMSE	PSNR	SAM	ERGAS	Q2n	UIQI	Runtimes(s)
GSA	0.0102	35.9799	3.5303	2.2934	0.9794	0.9805	1.0102
CNMF	0.0126	34.4203	2.7762	2.6042	0.9717	0.9684	13.1973
HySure	0.0056	41.9679	1.9931	1.1711	0.9751	0.9935	296.7299
FUSE	0.0119	34.8241	3.3454	2.4742	0.9844	0.9778	1.0796
CSTF	0.0051	42.4397	1.8157	1.0957	0.9754	0.9939	66.0898
LTMR	0.0053	42.2317	1.8372	1.1572	0.9748	0.9955	82.2640
STEREO	0.0065	40.2251	2.4507	1.3966	**0.9940**	0.9915	520.8580
BSCOTT	0.0063	40.1797	1.9919	1.4430	0.9735	0.9919	**0.7147**
IR-TenSR	0.0077	39.1613	2.7950	2.6238	0.9829	0.9830	157.2340
LTTR	0.0049	42.7337	1.7394	1.2352	0.9848	0.9939	342.7824
FDAssr	**0.0040**	**45.1006**	**1.0838**	**0.8323**	0.9853	**0.9980**	103.4848
Ideal value	0	$+\infty$	**0**	**0**	**1**	**1**	**0**

2) Visual Comparison: To further evaluate the quality of the fusion results, we conducted a visual comparison. We selected pseudo-color images composed of three bands and their corresponding absolute error maps, as well as magnified regions, to intuitively perceive the detailed information of the fusion results. The absolute error maps illustrate the color-based differences between the estimated images and the reference images, where blue indicates smaller errors. The visual comparison results for Pavia University, Chikusei, Houston, and Washington DC Mall are shown in Figs. 6–9, respectively. Across all four datasets, the HSI obtained using our method exhibited finer spatial details and absolute error maps closer to the ground truth (GT). This demonstrates the effectiveness of the proposed FDAssr method. Consistent with the quantitative results, the CSTF, LTMR, and LTTR methods also exhibited excellent visual performance. The GSA, CNMF, and FUSE methods exhibited poorer visual performance. This is likely due to their relatively simple modeling approaches and computational processes. The visual comparison clearly shows that our method has the smallest errors, highlighting the effectiveness of the proposed FDAssr method in the HSI-MSI fusion problem.

F. Experiments on Real Data Fusion

To further evaluate the performance of the proposed method (FDAssr) in real-world scenarios, we conducted comparative experiments on real data. The HypSen dataset [81] includes 30-m resolution HSI captured by the Hyperion imaging sensor on the Earth Observing-1 satellite and 10-m resolution MSI captured by the Sentinel-2A satellite for the same scene. After removing noisy bands and bands affected by water vapor absorption, we retained 66 bands for the LR-HSI and four bands for the HR-MSI. In the comparative experiments, we selected a 100×100 subimage from the LR-HSI and a corresponding 300×300 subimage from the HR-MSI. We used the proposed method to estimate the spatial and spectral degradation parameters and applied them to various fusion methods. The visual fusion results are shown in Fig. 10. Our method effectively reconstructs textures and edges in spatial details without losing color information. The IR-TenSR method lost some spatial detail, resulting in an overly smooth image, while the STEREO and BSCOTT methods lost spectral information, leading to color distortion. The visual perception comparison on real data further validates the superiority of our proposed method.

VI. ABLATION ANALYSIS

In this section, we further validate and analyze the effectiveness and rationale of our proposed method through ablation experiments.

A. Effectiveness of Spatial and Spectral Degradation Parameter Estimation

The main contribution of this study is the development of an efficient method for accurately estimating the spatial and spectral response parameters in the degradation model. These estimated parameters are then utilized to create a novel HSI-MSI fusion algorithm for generating high-resolution HSI. Previous research often overlooked the critical role of the degradation model in data fusion. In their simulated data experiments, predefined degradation parameters are typically used to generate the data, enabling the construction of fusion models that produce satisfactory results. However, in real-world scenarios, imaging systems are influenced by complex environmental factors, rendering the use of predefined parameters both inaccurate and theoretically unjustified for fusion. Therefore, it is essential to estimate the spatial and spectral response parameters accurately from observed data. In this section, we compare and evaluate the fusion results obtained using degradation parameters estimated through different approaches.

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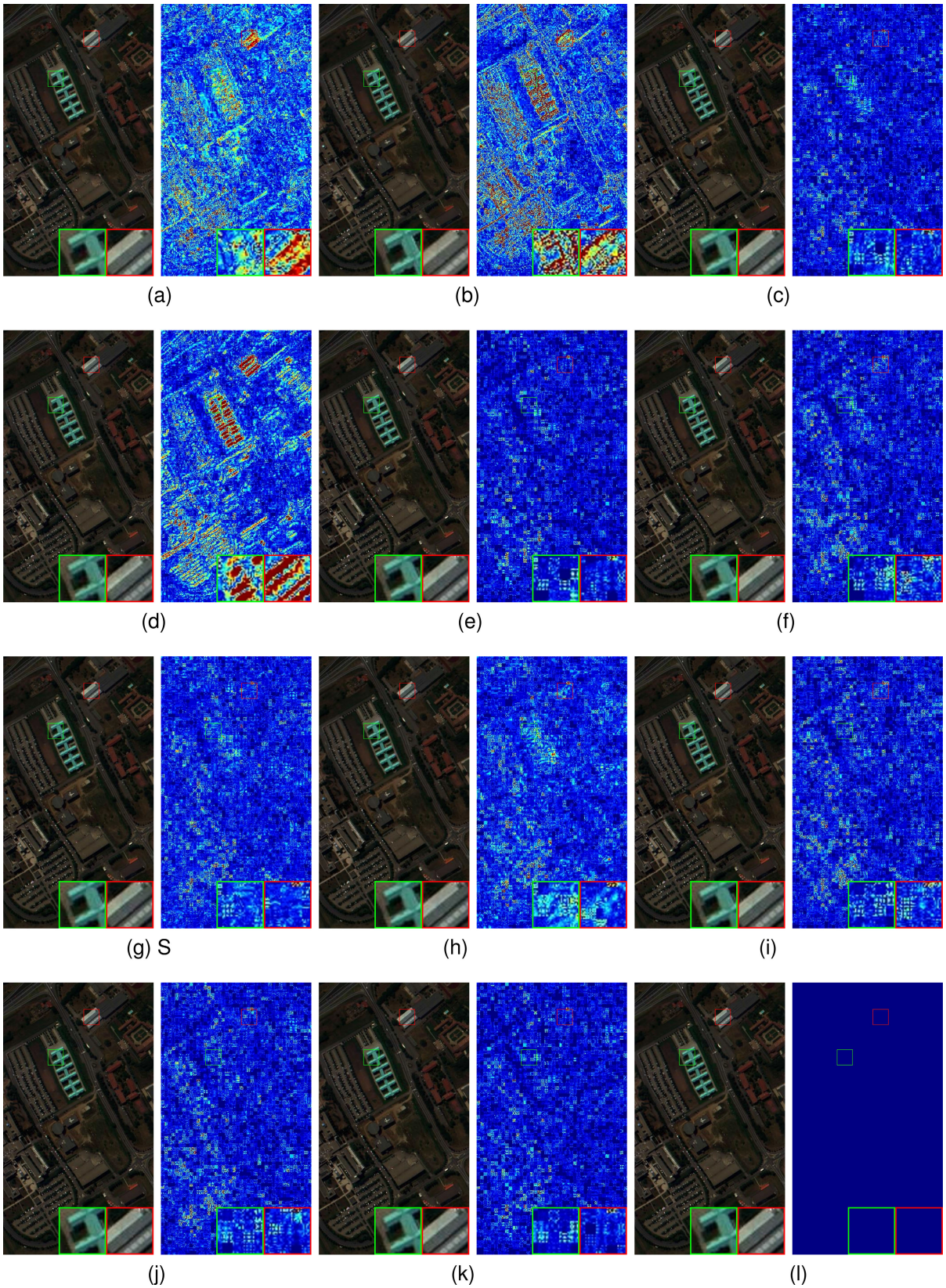


Fig. 6. Pseudo-color images formed by 60th, 29th, and seventh bands and the corresponding error images of fused Pavia University by test methods. (a) GSA [74]. (b) CNMF [52]. (c) HySure [46]. (d) FUSE [70]. (e) CSTF [75]. (f) LTMR [56]. (g) STEREO [61]. (h) BSCOTT [62]. (i) IR-TenSR [57]. (j) LTTR [58]. (k) FDAssr. (l) GT.

The real dataset, named Liao Ning-01 [39], contains HSI data acquired by the Chinese ZY-1 02D satellite over Dalian City, Liaoning Province. It consists of a $300 \times 300 \times 166$ LR-HSI and a corresponding $900 \times 900 \times 8$ HR-MSI. For

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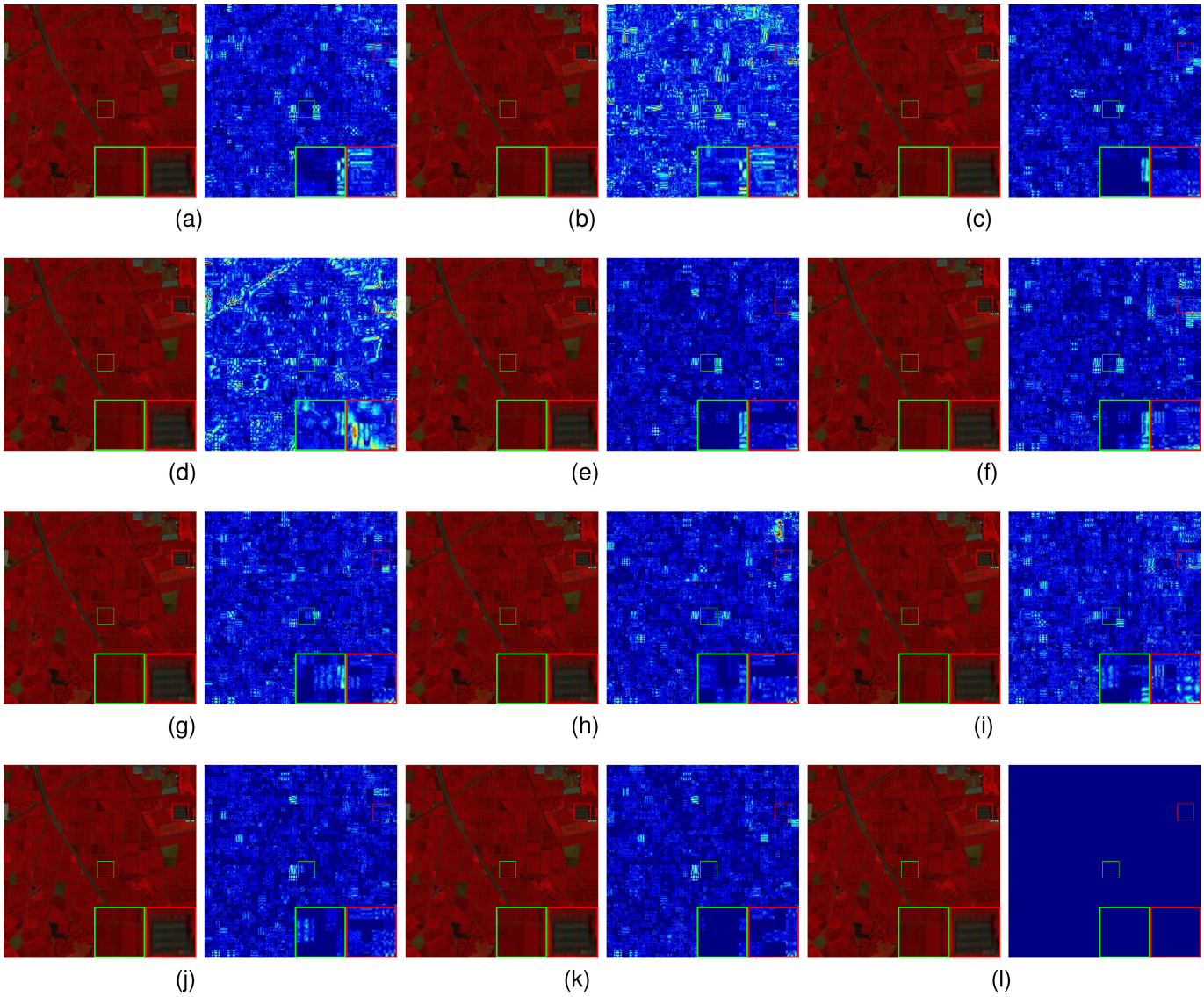


Fig. 7. Pseudo-color images formed by 100th, 60th, and tenth bands and the corresponding error images of fused Chikusei by test methods. (a) GSA [74]. (b) CNMF [52]. (c) HySure [46]. (d) FUSE [70]. (e) CSTF [75]. (f) LTMR [56]. (g) STEREO [61]. (h) BSCOTT [62]. (i) IR-TenSR [57]. (j) LTTR [58]. (k) FDAssr. (l) GT.

the ablation experiments, we used cropped subimages of size 100×100 and 300×300, respectively. In these experiments, we applied the spatial and spectral responses obtained through different degradation parameter estimation methods to our proposed fusion algorithm. Specifically, the fusion results using parameters estimated by the CNMF method are denoted as CNMF-SR, those estimated by the zero-shot learning (ZSL) method are denoted as ZSL-SR, and those estimated by our method are denoted as FDA-SR. Fig. 11 shows the different fusion results produced by the degradation parameters estimated using various methods. To evaluate performance, we calculated the mean spectral vector of all pixels. As shown in Fig. 11, the average spectral curve of the fusion using our estimated spatial and spectral responses is closer to the spectral curve of the original LR-HSI, which is represented as LrHSI and shown by the blue curve. This indicates that our estimated degradation parameters preserve more spectral information and are more suitable for the HSI-MSI fusion problem in real-world scenarios.

B. Effectiveness of Spatial–Spectral Regularization

In this section, we validate the effectiveness of the spatial content-aware SSR (26) through ablation experiments. Incorporating reasonable a priori information can effectively regularize the fusion model and enhance fusion accuracy. Note (29) consists of a data fidelity term and a regularization term. To demonstrate the impact of the proposed regularizer, we compare our FDAssr model with a simplified version that excludes the regularization term from (29) (i.e., setting $\mu = 0$), defined as follows:

$$\underset{\bar{\mathbf{B}}_{(3)}}{\operatorname{argmin}} \left\| \mathbf{X}_{(3)} - \mathbf{\Phi}\bar{\mathbf{B}}_{(3)}\mathbf{Q}\mathbf{S} \right\|_F^2 + \left\| \mathbf{Y}_{(3)} - \mathbf{R}\mathbf{\Phi}\bar{\mathbf{B}}_{(3)} \right\|_F^2. \tag{45}$$

We refer to the model (45) as the simplified model, which relies solely on the degradation model for estimation, denoted as FDAsimple in the experiments. The quantitative evaluation results on the four datasets are shown in Table VI. Our FDAssr method outperforms the FDAsimple in all quantitative evaluation metrics across all datasets. This indicates that,

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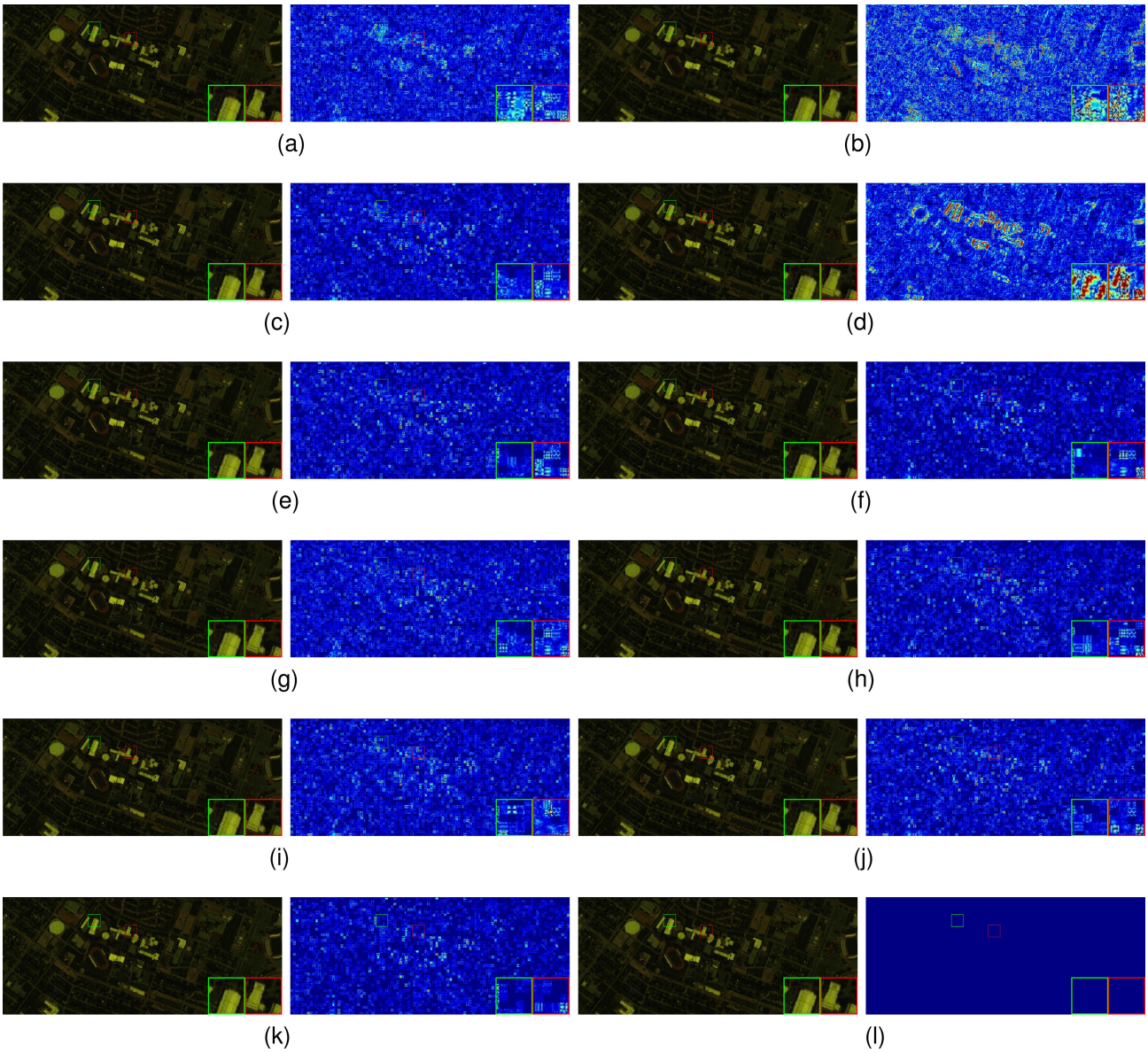


Fig. 8. Pseudo-color images formed by 60th, 29th, and seventh bands and the corresponding error images of fused Houston by test methods. (a) GSA [74]. (b) CNMF [52]. (c) HySure [46]. (d) FUSE [70]. (e) CSTF [75]. (f) LTMR [56]. (g) STEREO [61]. (h) BSCOTT [62]. (i) IR-TenSR [57]. (j) LTTR [58]. (k) FDAssr. (l) GT.

given accurately estimated spatial and spectral degradation parameters, the spatial content-aware SSR effectively enhances the estimation accuracy, resulting in superior fusion results.

C. Analysis of Sampling Factors

In this section, we conduct a comparative analysis of different sampling factors to validate the robustness of our method. Fig. 12 shows the HSI-MSI fusion results on the Washington DC Mall dataset under five different sampling factors. It is evident that as the sampling factor increases, the SAM and PSNR metrics for all methods gradually deteriorate. Although our proposed FDAssr is also affected, it consistently outperforms all comparison methods in most cases. Furthermore, Fig. 13 presents the absolute error maps of the fusion results under four different sampling factors. As the sampling factor increases, brighter areas appear in the error maps for each method, indicating more severe spatial information distortion. Each method is affected to varying degrees, but our method exhibits the smallest error under each sampling factor, further validating the superiority of our proposed FDAssr.

D. Analysis of Tuning Parameters

In this study, the fusion algorithm involves three parameters that need to be discussed: the number of superpixels T, the penalty parameter μ for the spatial content-aware SSR, and the parameter ρ used in ADMM optimization. The number of superpixels T is closely related to the complexity of the material distribution within the region. To balance computational efficiency and estimation accuracy, we set $T = 25$ in our experiments. The parameters μ and ρ have a significant influence over the solution of the fusion algorithm. Larger values indicate a stronger penalty. Fig. 14 illustrates the PSNR curves for the four different datasets as functions of μ and ρ. As shown in Fig. 14, the PSNR for all four datasets reaches its maximum when $\rho = 1e^{-3}$. The optimal value for μ varies slightly. For the Washington DC Mall dataset, the PSNR is

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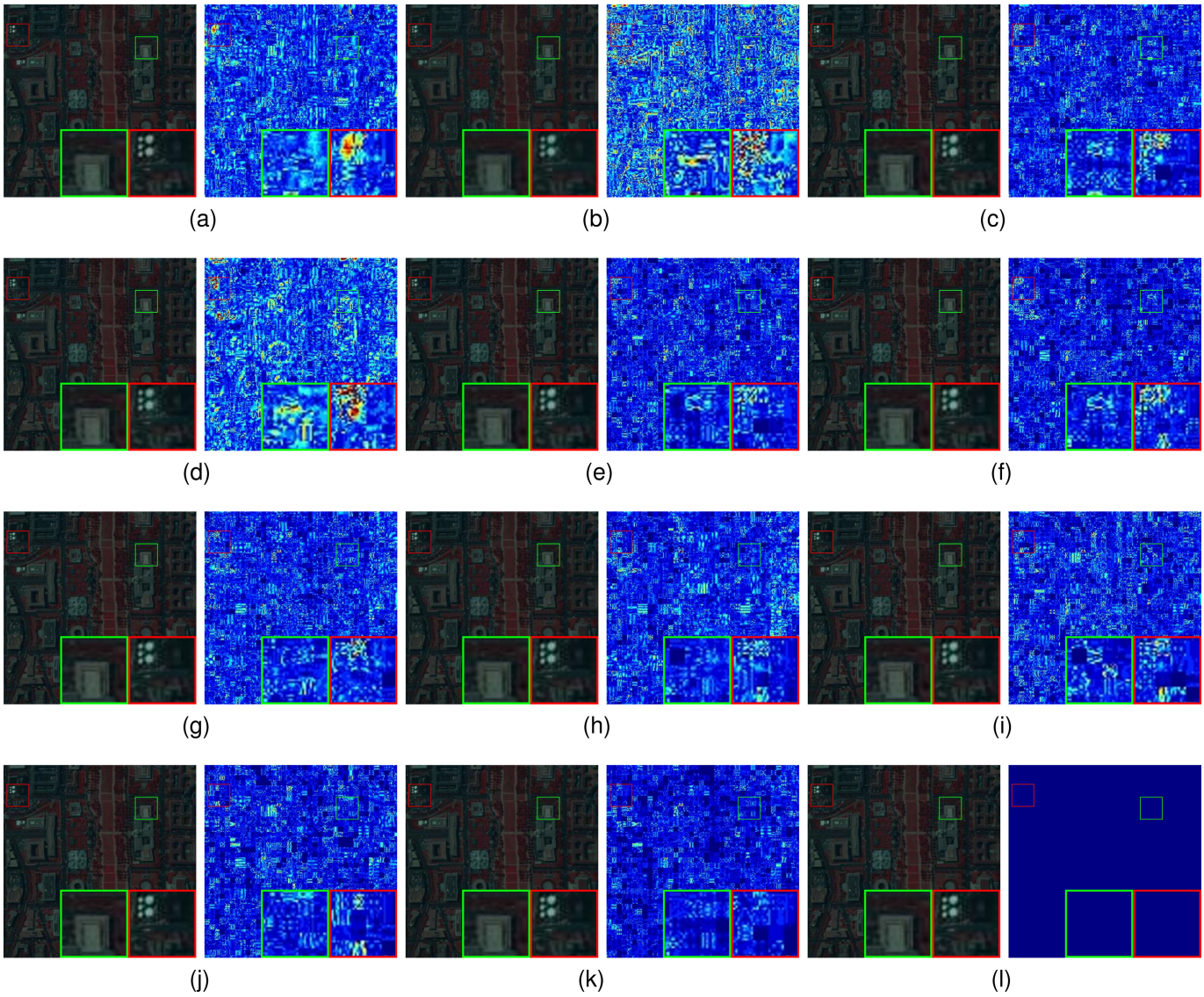


Fig. 9. Pseudo-color images formed by 60th, 27th, and 17th bands and the corresponding error images of fused Washington DC Mall by test methods. (a) GSA [74]. (b) CNMF [52]. (c) HySure [46]. (d) FUSE [70]. (e) CSTF [75]. (f) LTMR [56]. (g) STEREO [61]. (h) BSCOTT [62]. (i) IR-TenSR [57]. (j) LTTR [58]. (k) FDAssr. (l) GT.

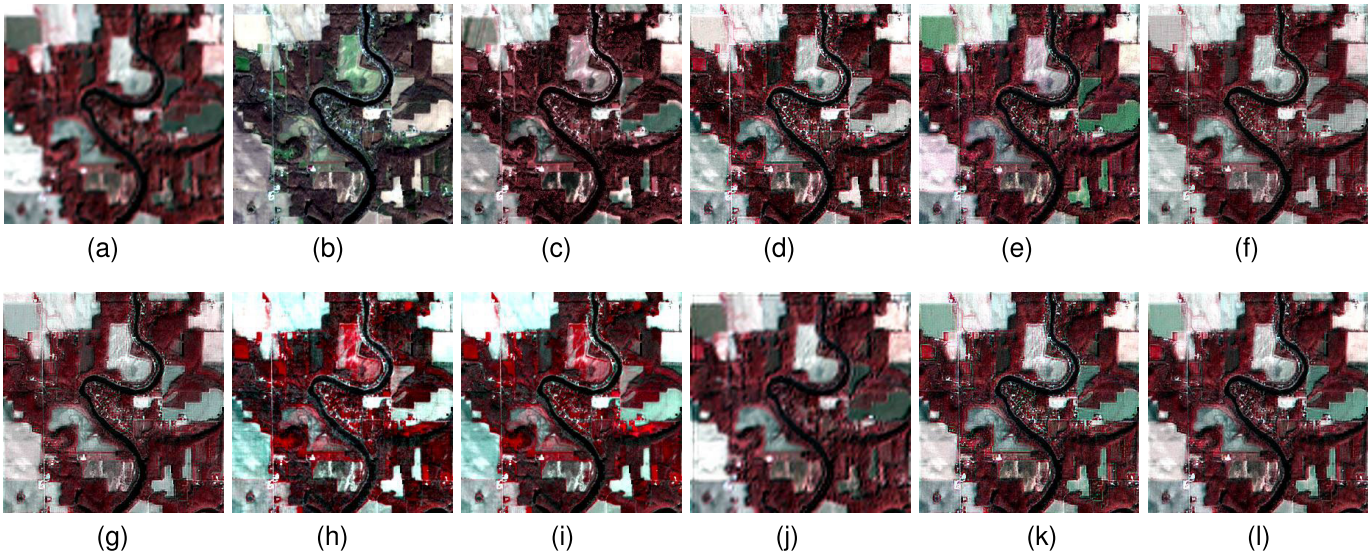


Fig. 10. Pseudo-color images formed by the 12th, fifth, and first bands of fused HypSen by the test methods. (a) LrHSI. (b) HrMSI. (c) CNMF [52]. (d) HySure [46]. (e) FUSE [70]. (f) CSTF [75]. (g) LTMR [56]. (h) STEREO [61]. (i) BSCOTT [62]. (j) IR-TenSR [57]. (k) LTTR [58]. (l) FDAssr.

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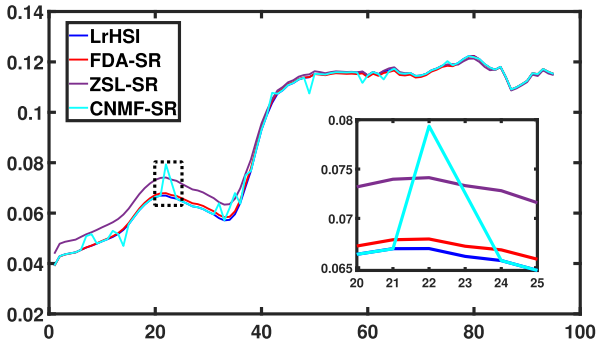


Fig. 11. Average spectral curves obtained from parameters estimated by different methods on the Liao Ning-01 dataset.

TABLE VI
OVERALL PERFORMANCE OF METRICS RMSE, PSNR, SAM, AND ERGAS BY OUR METHOD (WITH OR WITHOUT REGULARIZATION) ON FOUR DIFFERENT DATASETS

	RMSE	PSNR	SAM	ERGAS
Washington				
FDAsimple	0.0098	37.2438	3.9431	3.6616
FDAssr	0.0040	45.1006	1.0838	0.8323
Houston				
FDAsimple	0.0087	43.7929	3.7946	2.6991
FDAssr	0.0015	56.7172	0.6098	0.5723
Chikusei				
FDAsimple	0.0096	48.4365	1.7189	1.6582
FDAssr	0.0021	59.3623	0.4734	0.5766
PaviaU				
FDAsimple	0.0145	39.8734	4.0567	2.4643
FDAssr	0.0067	46.0234	2.1985	1.3468

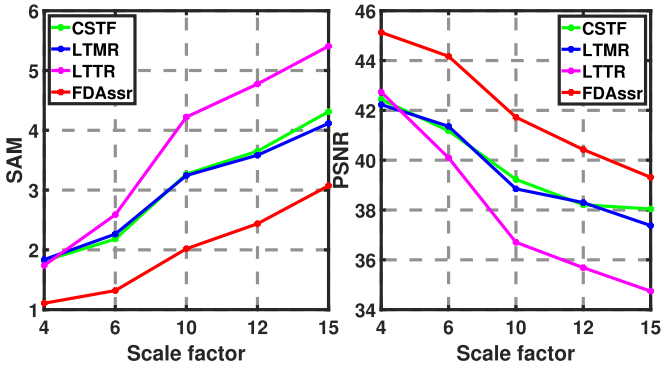


Fig. 12. SAM and PSNR metrics obtained with different scale factors on the Washington DC Mall dataset.

optimal when $\mu = 1e^{-3}$, while for the other three datasets, the PSNR is optimal when $\mu = 1e^{-4}$.

E. Analysis of the Initialization of Q

In this section, we constructed comparative experiments to examine the impact of the initial values of Q on the degradation parameter estimation. Specifically, we referred to the method of randomly generating initial values from a standard normal distribution for the elements of Q as "random," the method of setting all elements of Q to 1 as "ones," and the method of setting matrix Q as a Gaussian kernel with

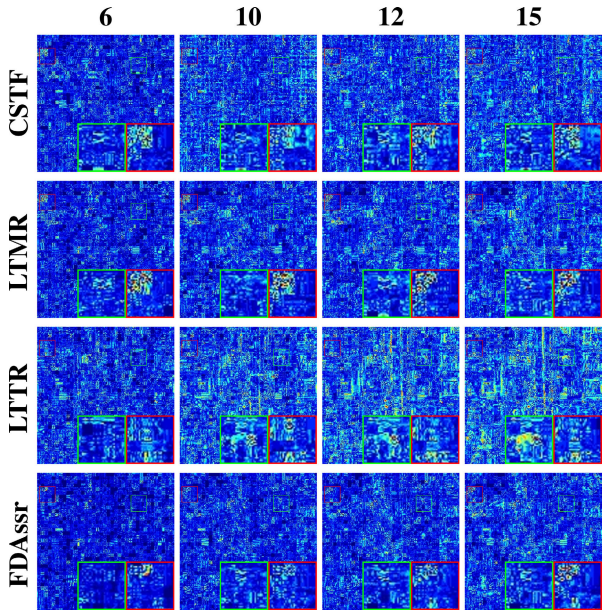


Fig. 13. Absolute error maps obtained with different scale factors on the Washington DC Mall dataset.

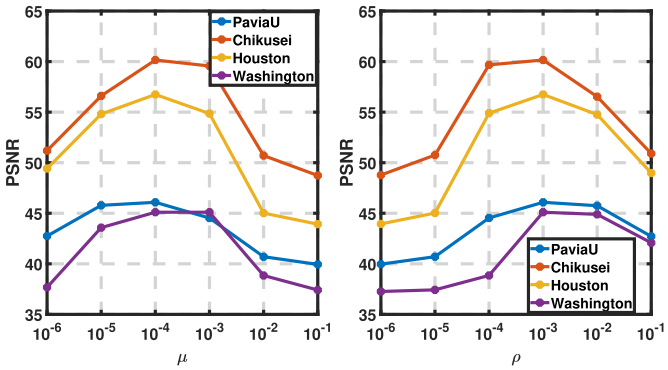


Fig. 14. PSNR values obtained for different parameter values of μ and ρ in four datasets.

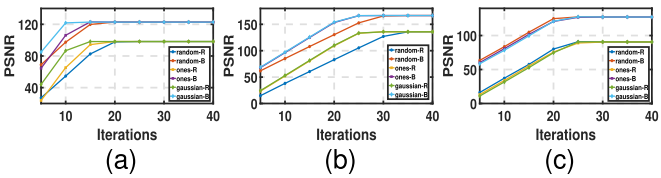


Fig. 15. PSNR values of the degradation parameter estimation results produced by Q with different initial states. (a) 5×5. (b) 7×7. (c) 9×9.

a standard deviation of 10 as "gaussian." We obtained the line chart shown in Fig. 15.

As shown in Fig. 15, under various kernel size, the initial values of Q only affect the convergence speed in the early iterations. However, when the iterations satisfy the convergence condition, the estimation results become stable and consistent. Therefore, the initial values of Q do not affect the final estimation of the degradation parameters.

VII. CONCLUSION AND FUTURE DIRECTION

In this article, we proposed a novel plug-and-play degradation parameter estimation method and utilized it to develop a new HSI-MSI fusion algorithm. This method can accurately capture the intrinsic features of different datasets. Specifically,

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we combined FDA techniques with an iterative optimization process to construct a new optimization algorithm for accurately estimating spatial and spectral response parameters. These parameters were then applied to the fusion algorithm to create a new HSI-MSI fusion algorithm. Additionally, we introduced and incorporated the spatial content-aware SSR, which efficiently leverages spatial and spectral low-rank characteristics to further enhance the fusion accuracy and efficiency.

In future work, we may consider using CNNs to estimate complex degradation parameters and combining them with optimization algorithms, which could potentially further improve the fusion accuracy.

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